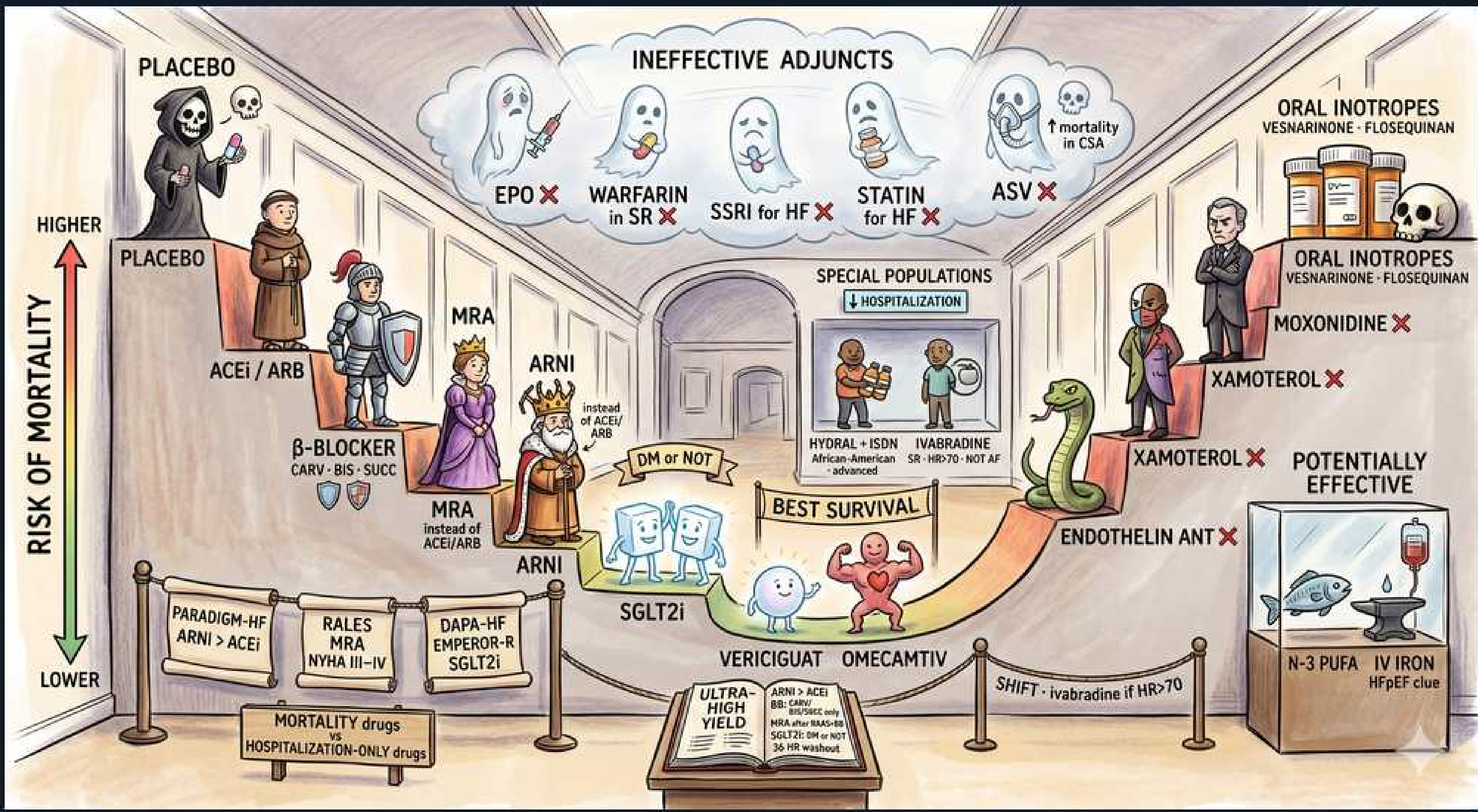
Heart Failure — SCENE 5 — HFrEF MORTALITY U-CURVE



1. High mortality end (placebo)

WHAT YOU SEE

Grim reaper + 'PLACEBO' at top-left, 'HIGHER risk of mortality'

WHAT IT ENCODES

Untreated HFrEF (placebo, the grim reaper) sits at the top of the curve — the highest-mortality reference point. Symptomatic HFrEF left untreated carries roughly 50% mortality at 5 years, comparable to many aggressive cancers, with the two dominant modes of death being sudden cardiac death (VT/VF) and progressive pump failure. Every proven drug class steps the patient DOWN this curve toward better survival.

BOARD TRAP

WRONG answer: 'HFrEF prognosis is benign once the patient is on a diuretic.' Discriminator: diuretics relieve congestion but do NOT reduce mortality; only the four pillars (ARNI, β-blocker, MRA, SGLT2i) move a patient down from the placebo/high-mortality end.

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2. ACEi / ARB step

WHAT YOU SEE

Pills 'ACEi / ARB' on the descending staircase

WHAT IT ENCODES

ACE inhibitors (e.g., enalapril target 10–20 mg BID, lisinopril) are the foundational RAAS-blocking step and the first historic mortality reducer in HFrEF (CONSENSUS, SOLVD). ARBs (candesartan, valsartan) are the alternative for ACEi-intolerant patients (cough, angioedema). They block angiotensin II–mediated vasoconstriction and remodeling. In modern algorithms this rung is superseded by ARNI when tolerated.

BOARD TRAP

WRONG answer: 'switch directly from an ACE inhibitor to an ARNI with no washout.' Discriminator: sacubitril/valsartan requires a ≥36-hour ACEi-free washout to avoid overlapping bradykinin effects and angioedema; ARBs need no washout but ACEi does.

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3. Beta-blocker step

WHAT YOU SEE

Knight 'β-BLOCKER CARV / BIS / SUCC'

WHAT IT ENCODES

Only three evidence-based β-blockers reduce mortality in HFrEF: carvedilol (target 25–50 mg BID), bisoprolol (target 10 mg daily), and metoprolol SUCCINATE (extended-release, target 200 mg daily). They blunt chronic sympathetic toxicity and reverse remodeling (COPERNICUS, CIBIS-II, MERIT-HF). Start low, go slow, and only when euvolemic — never initiate during acute decompensation. Mnemonic: the knight 'Carv/Bis/Succ.'

BOARD TRAP

WRONG answer: 'use metoprolol TARTRATE (or atenolol) for HFrEF mortality benefit.' Discriminator: only carvedilol, bisoprolol, and metoprolol SUCCINATE are proven; tartrate and other β-blockers lack mortality evidence and should not be substituted.

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4. MRA step

WHAT YOU SEE

Queen 'MRA'

WHAT IT ENCODES

Mineralocorticoid receptor antagonists — spironolactone (RALES, target 25–50 mg) and eplerenone (EPHESUS/EMPHASIS, target 50 mg) — add incremental mortality reduction by blocking aldosterone-driven fibrosis and remodeling. Indicated for NYHA II–IV with EF ≤35%. Require eGFR >30 and K+ <5.0; monitor potassium and creatinine. Eplerenone avoids spironolactone's gynecomastia.

BOARD TRAP

WRONG answer: 'start an MRA in a patient with K+ 5.6 and eGFR 25.' Discriminator: MRAs are contraindicated when K+ ≥5.0 or eGFR <30 due to life-threatening hyperkalemia; the discriminating step is checking potassium and renal function before initiating.

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5. ARNI > ACEi (PARADIGM-HF)

WHAT YOU SEE

'ARNI', sign 'Paradigm-HF: ARNI > ACEi'

WHAT IT ENCODES

ARNI = sacubitril/valsartan, an angiotensin-receptor–neprilysin inhibitor. PARADIGM-HF showed it was SUPERIOR to enalapril, cutting cardiovascular death and HF hospitalization by ~20%, so it REPLACES the ACEi/ARB rung rather than adding to it (target 97/103 mg BID). Neprilysin inhibition raises natriuretic peptides; this is why BNP becomes unreliable for monitoring while NT-proBNP stays valid.

BOARD TRAP

WRONG answer: 'add sacubitril/valsartan on top of the patient's lisinopril.' Discriminator: ARNI must REPLACE the ACEi (with a ≥36-h washout), never be combined with it — concurrent use causes dangerous angioedema risk.

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6. SGLT2i step (DAPA/EMPEROR)

WHAT YOU SEE

SGLT2i tablet, 'DAPA-HF / EMPEROR-R'

WHAT IT ENCODES

SGLT2 inhibitors — dapagliflozin (DAPA-HF, 10 mg) and empagliflozin (EMPEROR-Reduced, 10 mg) — are the 4th pillar, reducing CV death and HF hospitalization REGARDLESS of diabetes status. They sit at the lowest part of the curve and act via natriuresis, improved cardiac energetics, and reduced preload/afterload. Watch for genital mycotic infections and euglycemic DKA.

BOARD TRAP

WRONG answer: 'SGLT2 inhibitors only help heart failure in patients who have diabetes.' Discriminator: DAPA-HF and EMPEROR-Reduced proved benefit independent of diabetes; withholding an SGLT2i because the patient is non-diabetic is the trap.

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7. Best survival (four pillars)

WHAT YOU SEE

Strong figure 'BEST SURVIVAL' at the curve's bottom

WHAT IT ENCODES

Combined four-pillar GDMT — ARNI + evidence-based β-blocker + MRA + SGLT2i — produces the LOWEST mortality and best survival, and the gains are additive across pillars. Modern practice favors rapid SEQUENTIAL or simultaneous initiation of all four (low doses first, then titrate) rather than the old slow stepwise approach, because each pillar independently saves lives within weeks. This is the 'best survival' strongman at the curve's bottom.

BOARD TRAP

WRONG answer: 'fully up-titrate one drug class to maximum before introducing the next pillar.' Discriminator: current evidence favors getting all four pillars on board early at low doses, then titrating; sequential full-dose loading delays mortality benefit.

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8. Vericiguat / omecamtiv

WHAT YOU SEE

Open book 'VERICIGUAT / OMECAMTIV — ultra-high yield'

WHAT IT ENCODES

Newer adjuncts beyond the four pillars: vericiguat (soluble guanylate cyclase stimulator, VICTORIA) and omecamtiv mecarbil (selective cardiac myosin activator, GALACTIC-HF). Both modestly reduce HF HOSPITALIZATION or the composite endpoint in advanced/worsening HFrEF but do NOT substantially reduce all-cause mortality. They are add-ons for high-risk patients already on GDMT, not replacements for the pillars.

BOARD TRAP

WRONG answer: 'add vericiguat or omecamtiv to strongly reduce mortality.' Discriminator: their proven benefit is on hospitalization/composite endpoints, not a substantial mortality reduction — don't pick them as primary mortality-lowering agents over the four pillars.

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9. Ivabandine (HR ≥70, sinus)

WHAT YOU SEE

'HYDRAL + ISDN' and 'IVABRADINE SR >70' special-population box

WHAT IT ENCODES

Special-population add-ons. Ivabradine (SHIFT trial) inhibits the SA-node funny (If) current, lowering heart rate with no effect on contractility; add it when the patient is in SINUS rhythm with resting HR ≥70 despite a maximally tolerated β-blocker. Hydralazine + isosorbide dinitrate (A-HeFT) reduces mortality, particularly in self-identified African-American patients on GDMT, and is the go-to vasodilator when ACEi/ARB/ARNI are contraindicated (renal failure, hyperkalemia).

BOARD TRAP

WRONG answer: 'start ivabradine to control rate in an HFrEF patient who is in atrial fibrillation.' Discriminator: ivabradine works ONLY in sinus rhythm (it acts on the SA node); it is ineffective in AF and is indicated only when HR ≥70 persists despite optimal β-blockade.

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10. Ineffective: EPO, warfarin, SSRI, statin, ASV

WHAT YOU SEE

Ghosts: EPO, WARFARIN in SR, SSRI for HF, STATIN for HF, ASV (↑mortality in CSA)

WHAT IT ENCODES

The 'ineffective adjuncts' ghosts: erythropoietin for HF-related anemia (RED-HF — no benefit, raised thromboembolism), routine warfarin/low-dose anticoagulation in sinus rhythm without another indication (WARCEF — no net benefit), SSRIs for depression in HF (no outcome benefit), and statins for HF itself (CORONA/GISSI-HF — neutral, though still indicated for coexisting ASCVD). Adaptive servo-ventilation (ASV) is worse than useless — SERVE-HF showed it INCREASES mortality in HFrEF with central sleep apnea.

BOARD TRAP

WRONG answer: 'start a statin to improve survival in this HFrEF patient,' or 'use ASV for his central sleep apnea.' Discriminator: statins are neutral for HF itself (treat ASCVD, not HF), and ASV is HARMFUL (increases mortality) in HFrEF with central sleep apnea — actively contraindicated.

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11. Oral inotropes □ (harmful)

WHAT YOU SEE

Pill bottles 'ORAL INOTROPES vesnarinone / flosequinan' with □

WHAT IT ENCODES

Chronic ORAL inotropes are on the harm (rising) limb of the curve: vesnarinone, flosequinan, and long-term oral milrinone (PROMISE) all INCREASE mortality by driving arrhythmias and myocardial oxygen demand. IV inotropes (dobutamine, milrinone) are reserved for short-term use in cardiogenic shock or as a palliative bridge — never as chronic oral therapy in ambulatory HFrEF.

BOARD TRAP

WRONG answer: 'prescribe a chronic oral inotrope to improve contractility and survival in stable HFrEF.' Discriminator: long-term oral inotropes increase mortality; their only legitimate roles are short-term IV support in shock or palliation/bridging — not chronic outpatient use.

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12. Xamoterol / moxonidine □

WHAT YOU SEE

'MOXONIDINE □', 'XAMOTEROL □'

WHAT IT ENCODES

Two more harm-limb drugs. Moxonidine, a centrally-acting imidazoline/sympatholytic, INCREASED mortality in MOXCON. Xamoterol, a partial β-agonist, also increased mortality. Both illustrate that the wrong kind of neurohormonal manipulation (partial agonism or central sympatholysis) is dangerous — contrast with the proven mortality benefit of full β-blockade and RAAS inhibition.

BOARD TRAP

WRONG answer: 'a partial β-agonist like xamoterol can provide gentle inotropic support without the harm of full agonists.' Discriminator: xamoterol (and moxonidine) increased mortality in trials; partial β-agonism is harmful in HFrEF, the opposite of evidence-based β-blockade.

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13. Endothelin antagonist □

WHAT YOU SEE

Snake 'ENDOTHELIN ANT □'

WHAT IT ENCODES

Endothelin receptor antagonists (bosentan, darusentan), despite endothelin's potent vasoconstrictor role in HF, FAILED to improve outcomes and caused fluid retention and worsening HF in trials. They have no role in HFrEF (the snake on the harm side). Reserve endothelin antagonists for pulmonary arterial hypertension, not left-heart failure.

BOARD TRAP

WRONG answer: 'block the elevated endothelin levels in HFrEF with bosentan to improve outcomes.' Discriminator: endothelin antagonists worsen HF (fluid retention) with no survival benefit; they belong in PAH management, not HFrEF.

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14. Potentially effective: n-3 PUFA, IV iron

WHAT YOU SEE

Fish-oil & IV bag 'N-3 PUFA □ IV IRON □ (HFpEF clue)'

WHAT IT ENCODES

Modestly/potentially effective adjuncts: n-3 PUFA (omega-3 fish oil, GISSI-HF) gives a small reduction in death/hospitalization. IV iron — ferric carboxymaltose (FAIR-HF, AFFIRM-AHF) — corrects iron deficiency (ferritin <100, or 100–300 with TSAT <20%) to improve symptoms, exercise capacity, and QOL and reduce hospitalizations, even WITHOUT anemia. Use IV iron, not oral, in HF (poor GI absorption).

BOARD TRAP

WRONG answer: 'treat this iron-deficient HF patient with oral ferrous sulfate.' Discriminator: in HFrEF, IV iron (ferric carboxymaltose) is the evidence-based route — oral iron (IRONOUT-HF) failed to replete stores or improve outcomes; and iron repletion is indicated even when hemoglobin is normal.
